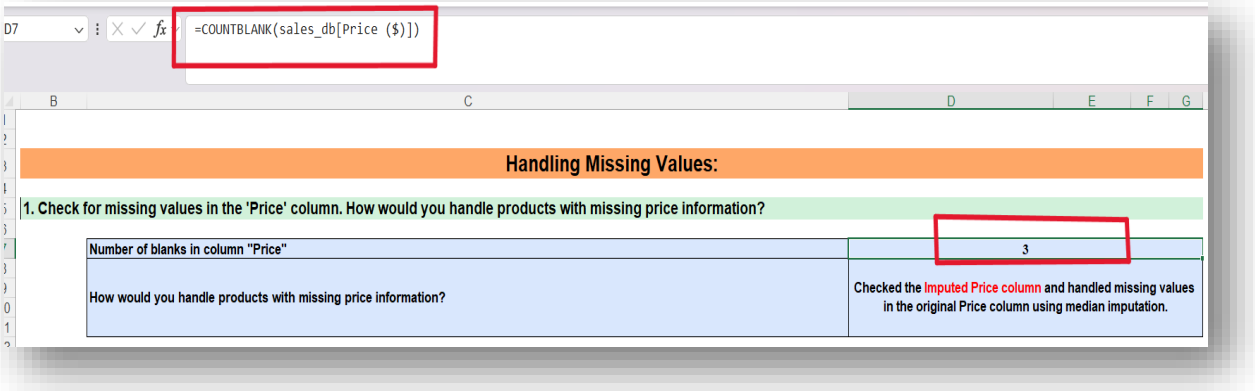
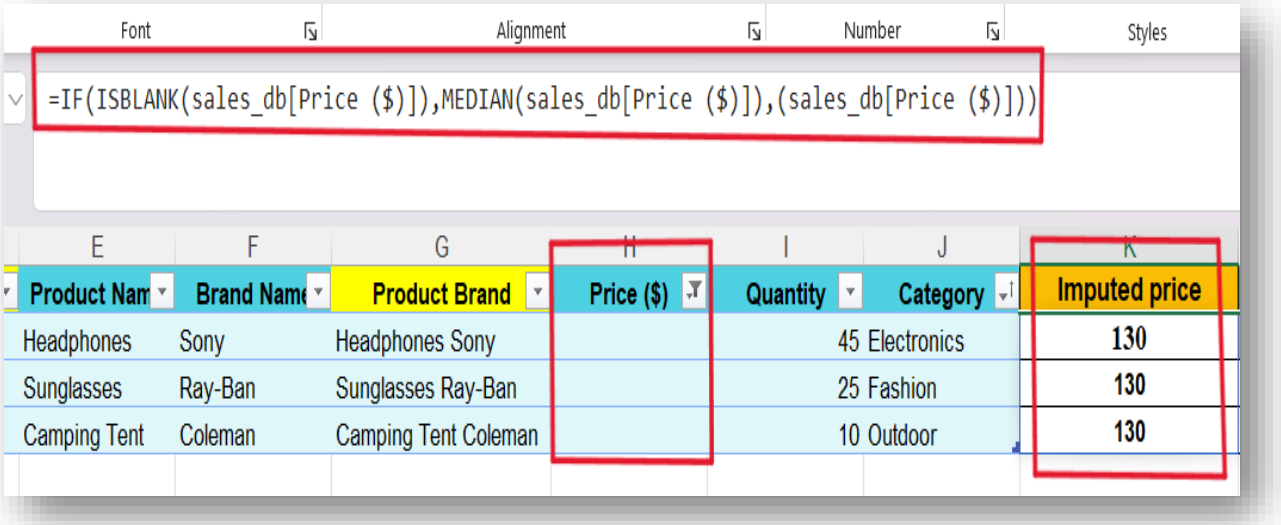
# DATA CLEANING AND TRANSFORMATION

## Handling Missing Values:

**1. Check for missing values in the 'Price' column. How would you handle products with missing price information?**

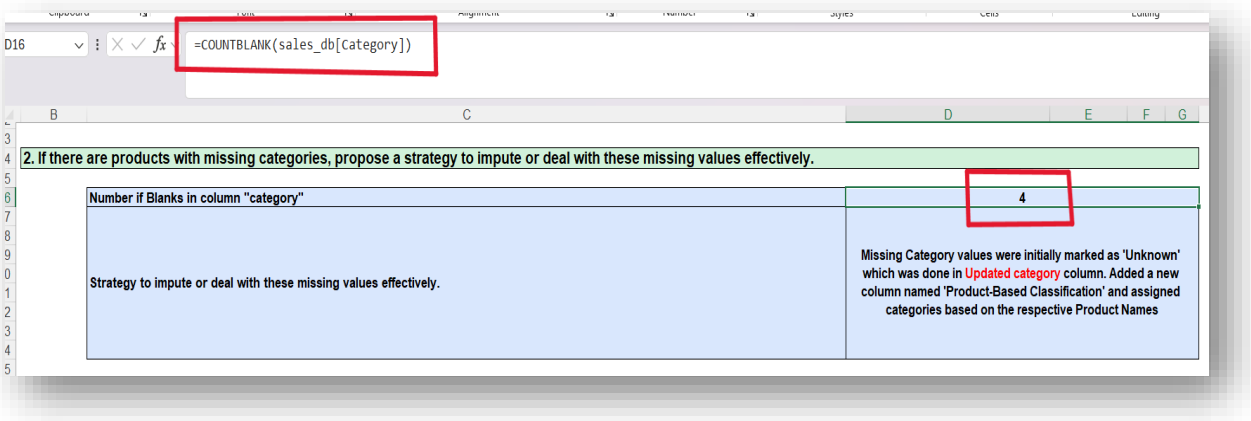
**Checked the Imputed Price column and handled missing values in the original Price column using median imputation.**

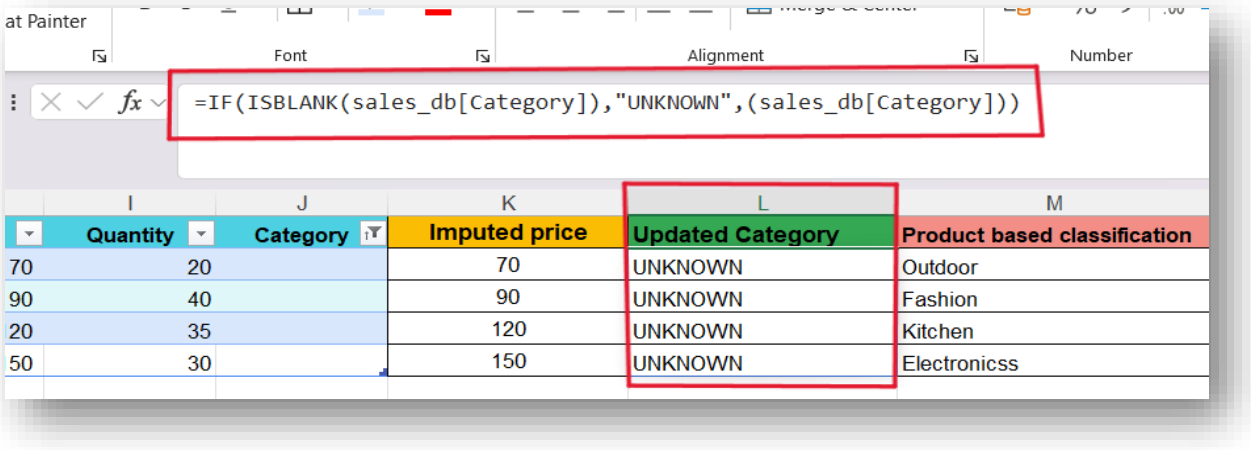


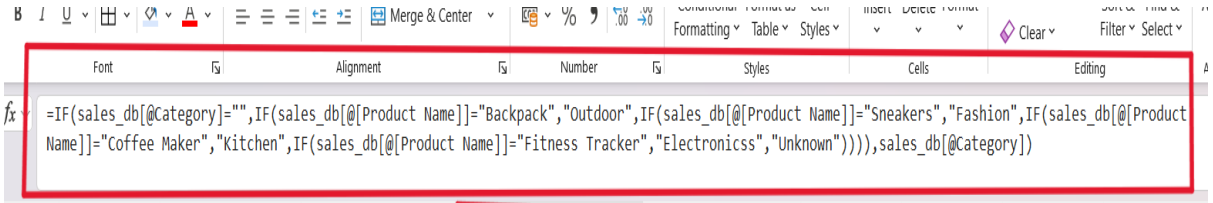
**2. If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.**

**Missing Category values were initially marked as 'Unknown' which was done in Updated category column. Added a new column named 'Product-Based Classification' and assigned categories based on the respective Product Names.**




| Quantity | Category | Imputed price | Updated Category | Product based classification |
|---|---|---|---|---|
| 70 | 20 | 70 | UNKNOWN | Outdoor |
| 90 | 40 | 90 | UNKNOWN | Fashion |
| 20 | 35 | 120 | UNKNOWN | Kitchen |
| 50 | 30 | 150 | UNKNOWN | Electronicss |

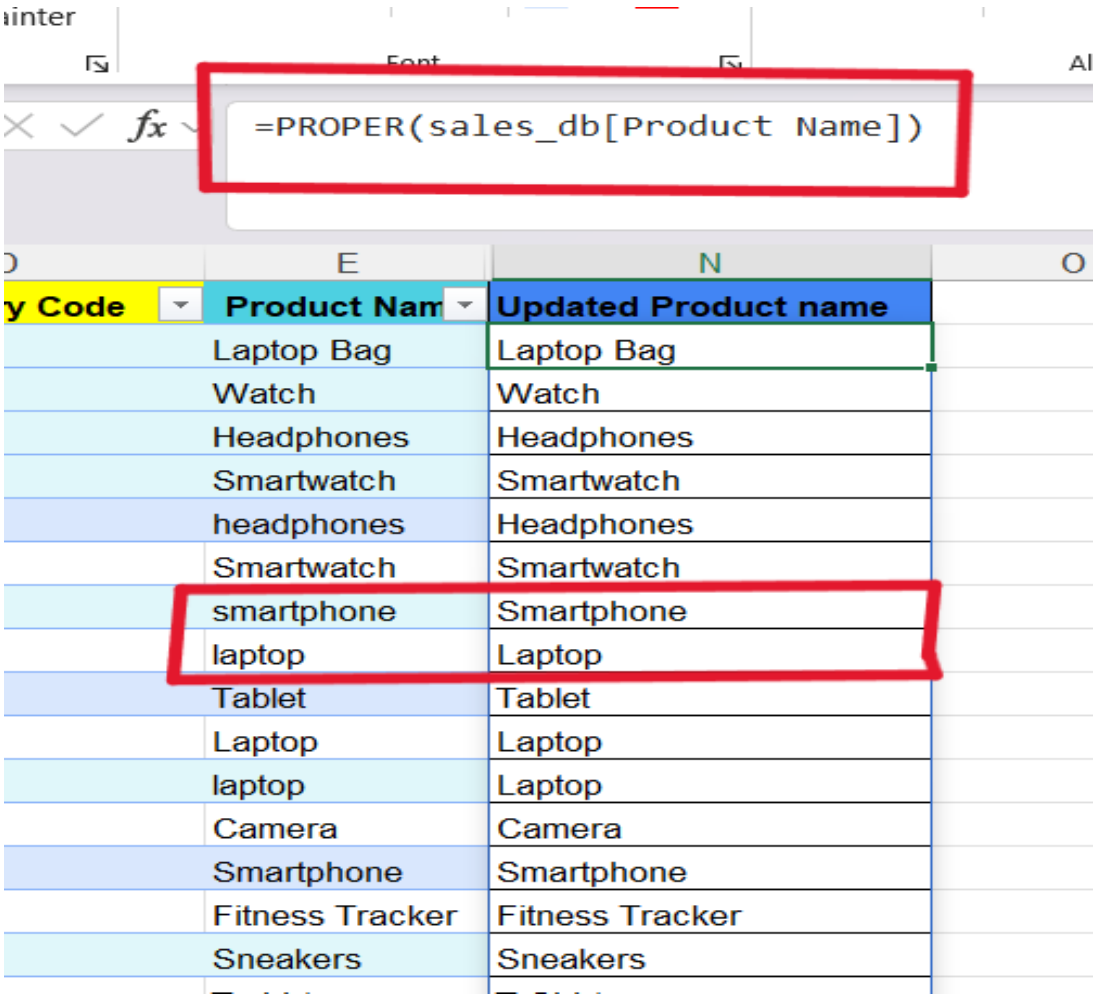



| Category | Imputed price | Updated Category | Product based classification | Updated Product name |
|---|---|---|---|---|
| | 70 | UNKNOWN | Outdoor | Backpack |
| | 90 | UNKNOWN | Fashion | Sneakers |
| | 120 | UNKNOWN | Kitchen | Coffee Maker |
| | 150 | UNKNOWN | Electronicss | Fitness Tracker |

# Correcting Inconsistent Data:

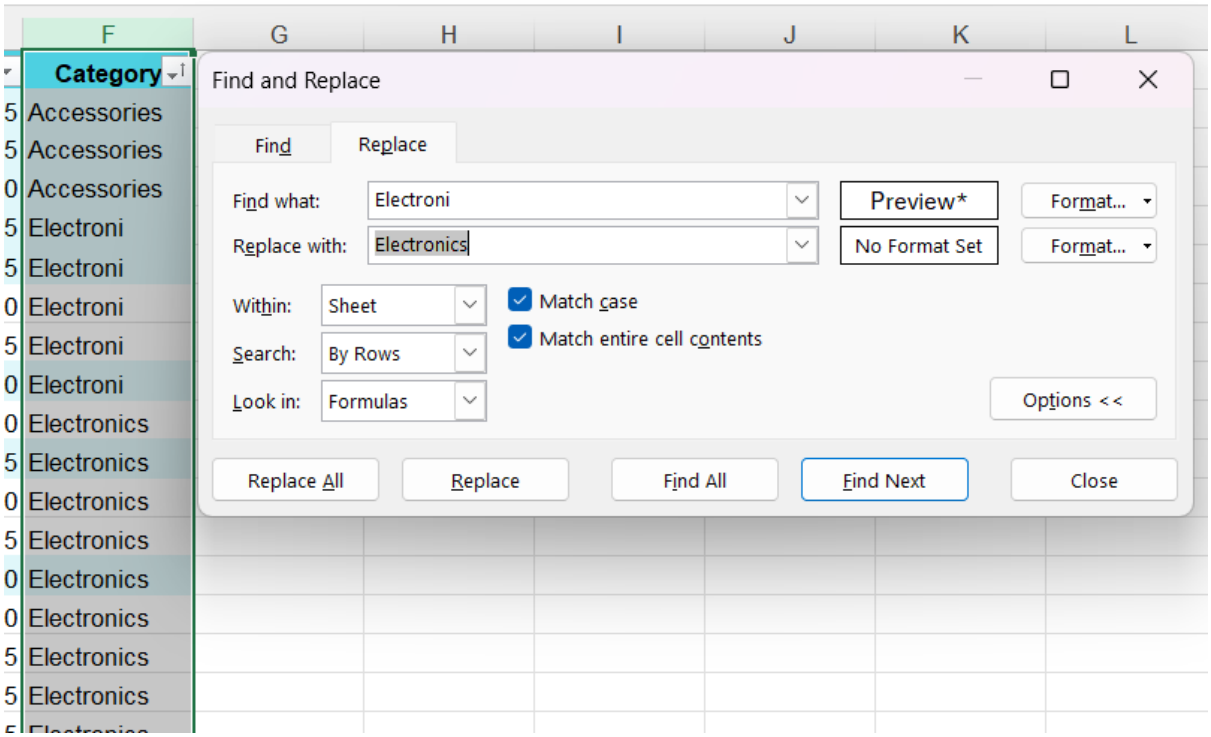
**1.Identify any inconsistent text formats present in the "Product Name" column.**

**Used the PROPER function to correct the text formatting in the "Updated Product Name "column**



**2. Identify any typos present in the "Category" column.**

**Sorted the Category column from A to Z, identified spelling mistakes (Electroni), and corrected them to Electronics using Find & Replace. This was done in "Category column" itself.**

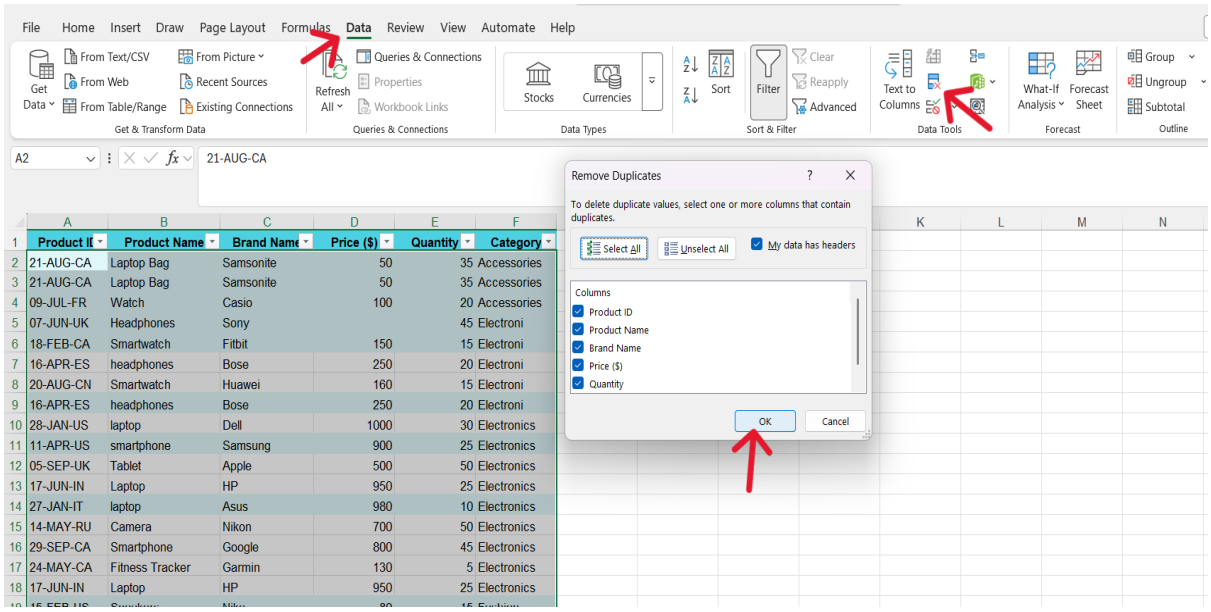


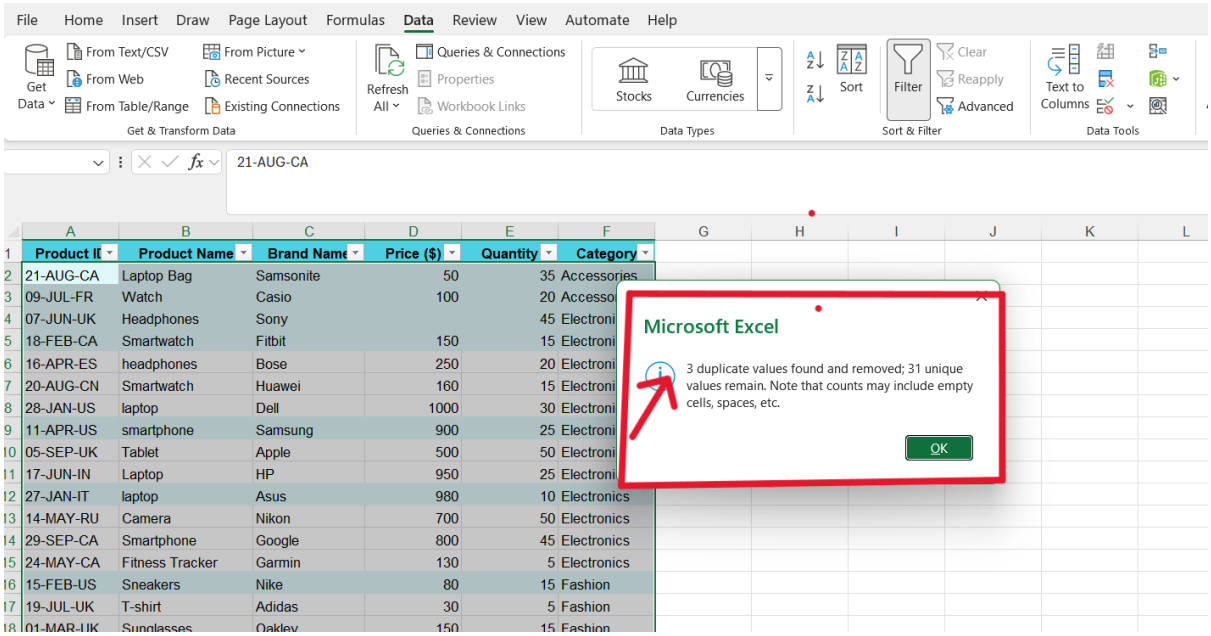
# Removing Duplicates:

**1. Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.**

**Removed duplicate records using the Remove Duplicates option from the Data tab.**

**3 duplicates have been removed, and 31 unique values has been remained.**

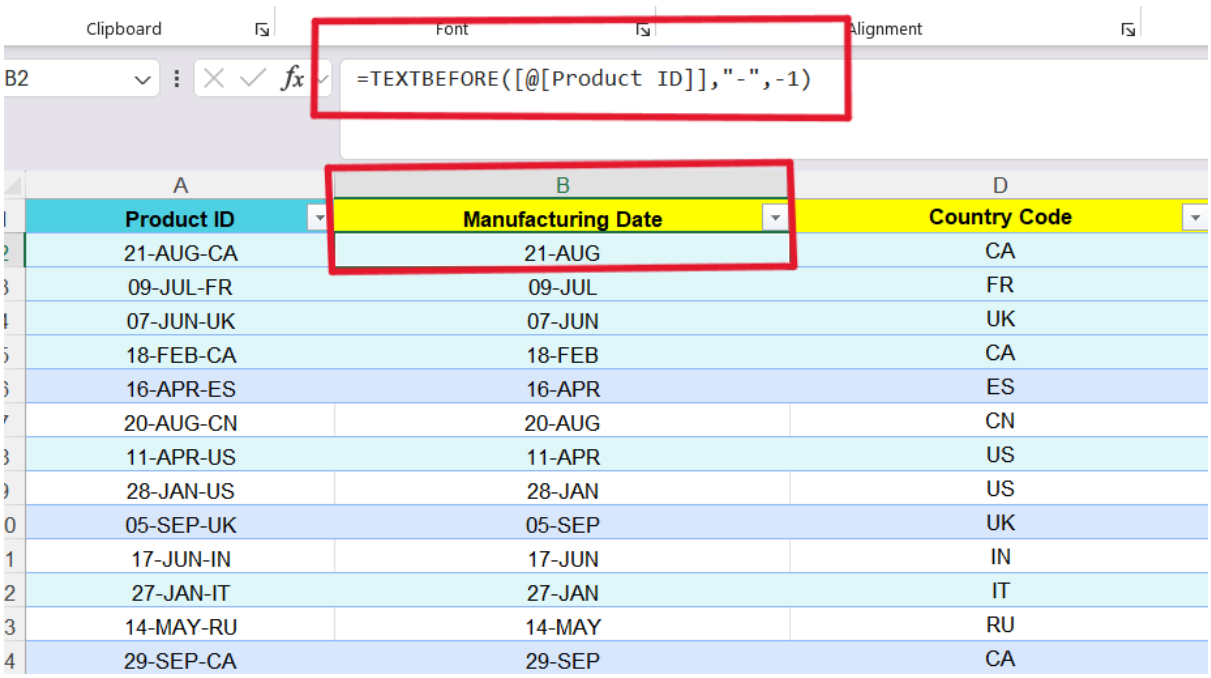


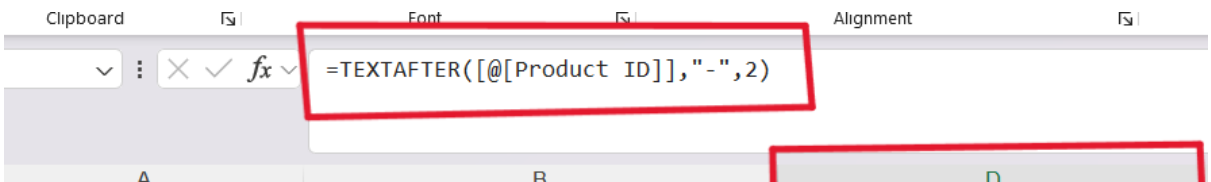
## Splitting and Merging Data:

**1. Split the "Product ID" column into two separate columns for " Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.**

**Used the TEXTBEFORE and TEXTAFTER functions to split the Product ID into separate columns. Added new column called "manufacturing date" and "Country code"**



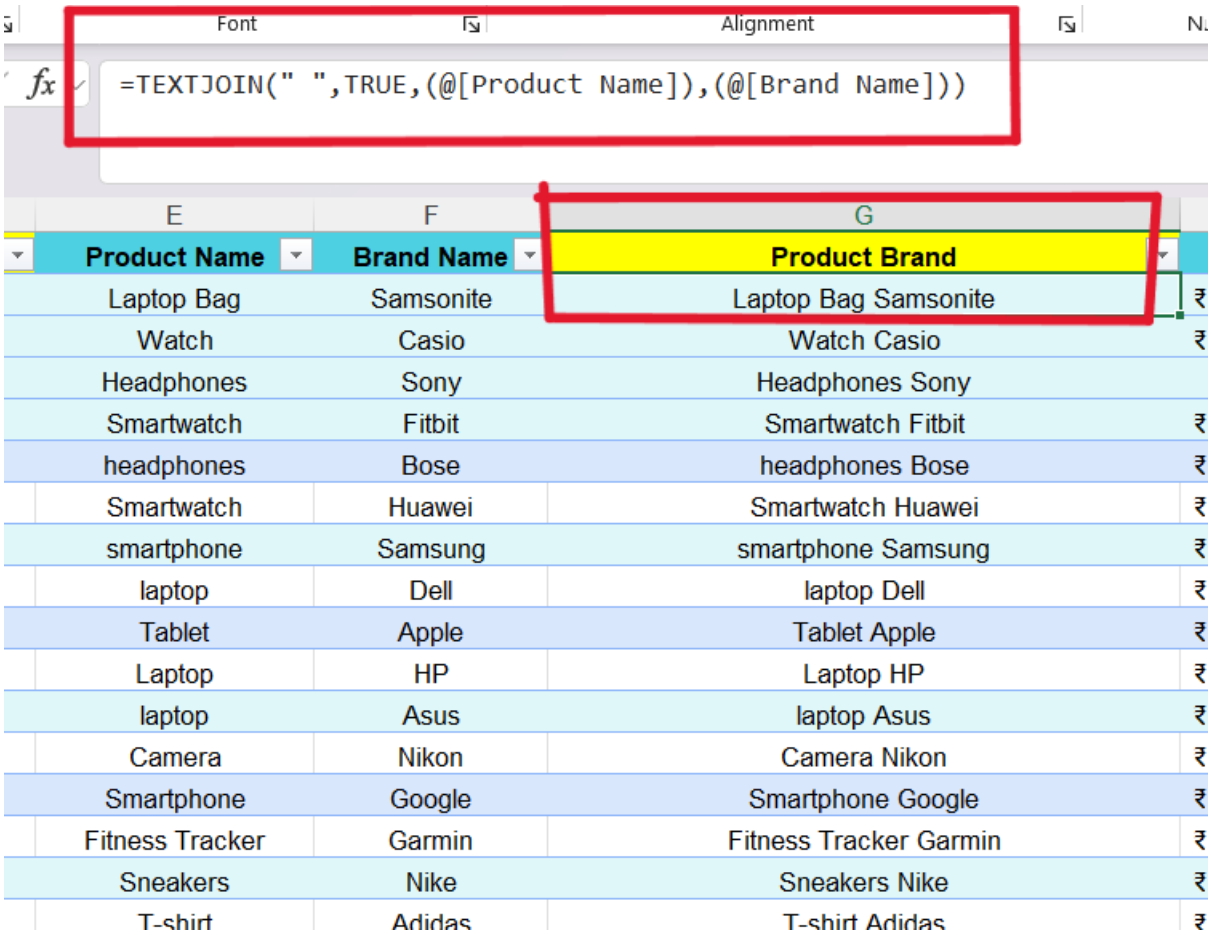

| | A | B | D |
|---|---|---|---|
| | Product ID | Manufacturing Date | Country Code |
| 2 | 21-AUG-CA | 21-AUG | CA |
| 3 | 09-JUL-FR | 09-JUL | FR |
| 4 | 07-JUN-UK | 07-JUN | UK |
| 5 | 18-FEB-CA | 18-FEB | CA |
| 6 | 16-APR-ES | 16-APR | ES |
| 7 | 20-AUG-CN | 20-AUG | CN |
| 8 | 11-APR-US | 11-APR | US |
| 9 | 28-JAN-US | 28-JAN | US |
| 10 | 05-SEP-UK | 05-SEP | UK |
| 11 | 17-JUN-IN | 17-JUN | IN |
| 12 | 27-JAN-IT | 27-JAN | IT |
| 13 | 14-MAY-RU | 14-MAY | RU |
| 14 | 29-SEP-CA | 29-SEP | CA |




| A | B | D |
|---|---|---|
| Product ID | Manufacturing Date | Country Code |
| 21-AUG-CA | 21-AUG | CA |
| 09-JUL-FR | 09-JUL | FR |
| 07-JUN-UK | 07-JUN | UK |
| 18-FEB-CA | 18-FEB | CA |
| 16-APR-ES | 16-APR | ES |
| 20-AUG-CN | 20-AUG | CN |
| 11-APR-US | 11-APR | US |
| 28-JAN-US | 28-JAN | US |
| 05-SEP-UK | 05-SEP | UK |
| 17-JUN-IN | 17-JUN | IN |
| 27-JAN-IT | 27-JAN | IT |
| 14-MAY-RU | 14-MAY | RU |

**2. Merge the "Brand Name" and "Product Name" columns into one column named "Product Brand".**

**Used the TEXTJOIN function to combine the Brand Name and Product Name columns into one column named "Product Brand".**

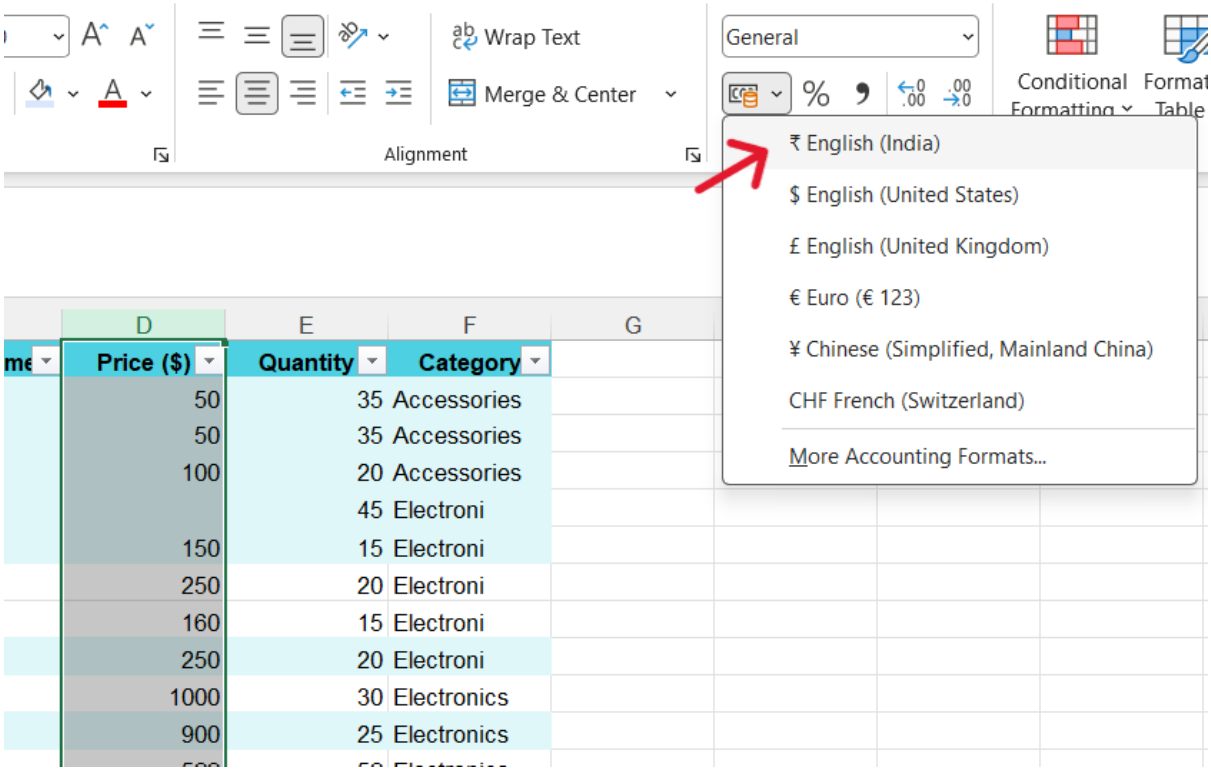


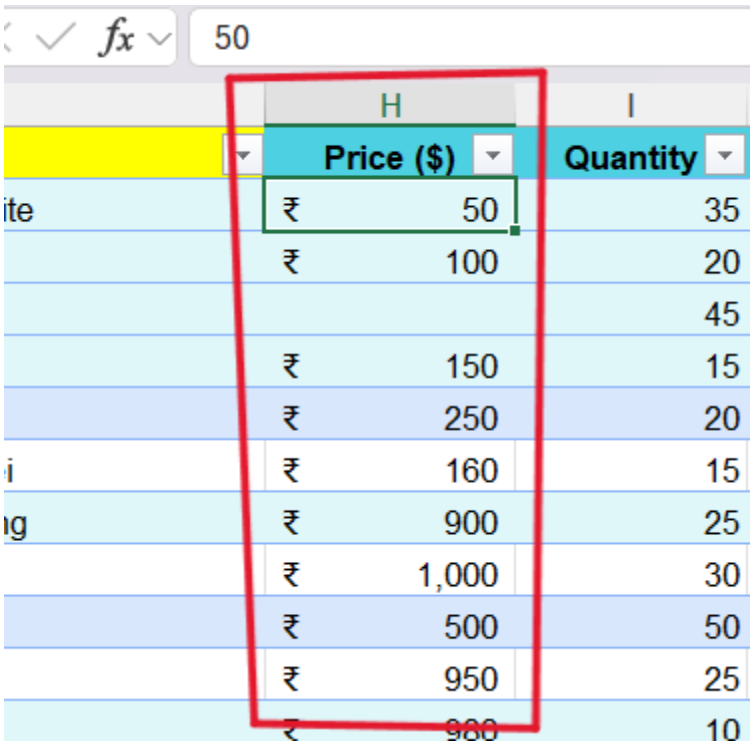

| E | F | G |
|---|---|---|
| Product Name | Brand Name | Product Brand |
| Laptop Bag | Samsonite | Laptop Bag Samsonite |
| Watch | Casio | Watch Casio |
| Headphones | Sony | Headphones Sony |
| Smartwatch | Fitbit | Smartwatch Fitbit |
| headphones | Bose | headphones Bose |
| Smartwatch | Huawei | Smartwatch Huawei |
| smartphone | Samsung | smartphone Samsung |
| laptop | Dell | laptop Dell |
| Tablet | Apple | Tablet Apple |
| Laptop | HP | Laptop HP |
| laptop | Asus | laptop Asus |
| Camera | Nikon | Camera Nikon |
| Smartphone | Google | Smartphone Google |
| Fitness Tracker | Garmin | Fitness Tracker Garmin |
| Sneakers | Nike | Sneakers Nike |
| T-shirt | Adidas | T-shirt Adidas |

# Number Formatting:

**1. Format the data type of the "Price" column to a currency format.**

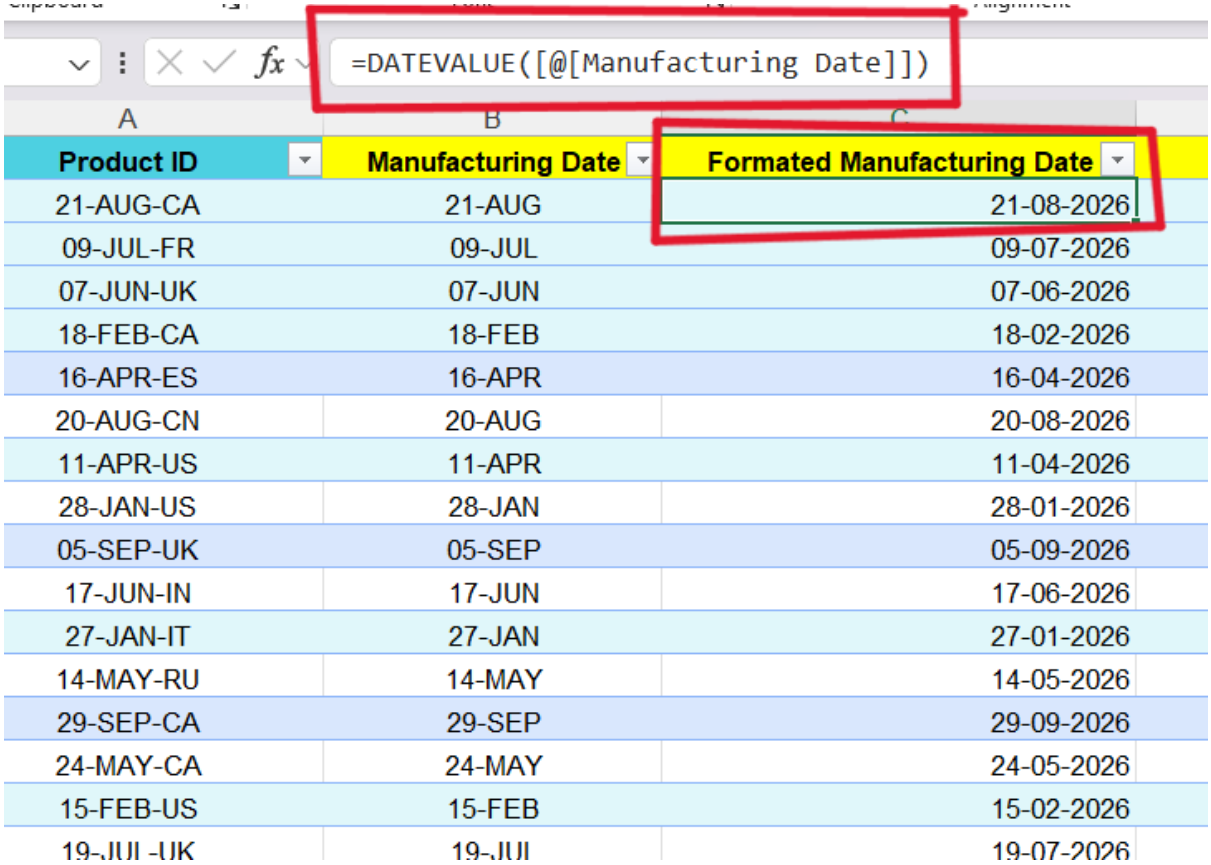
**Applied the Currency format to the Price column using the Currency option in the Home ribbon.**



**2. Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY " format.**

**Used the DATEVALUE function to convert text into a date format. This was done in the column "Formatted manufacturing date" column.**

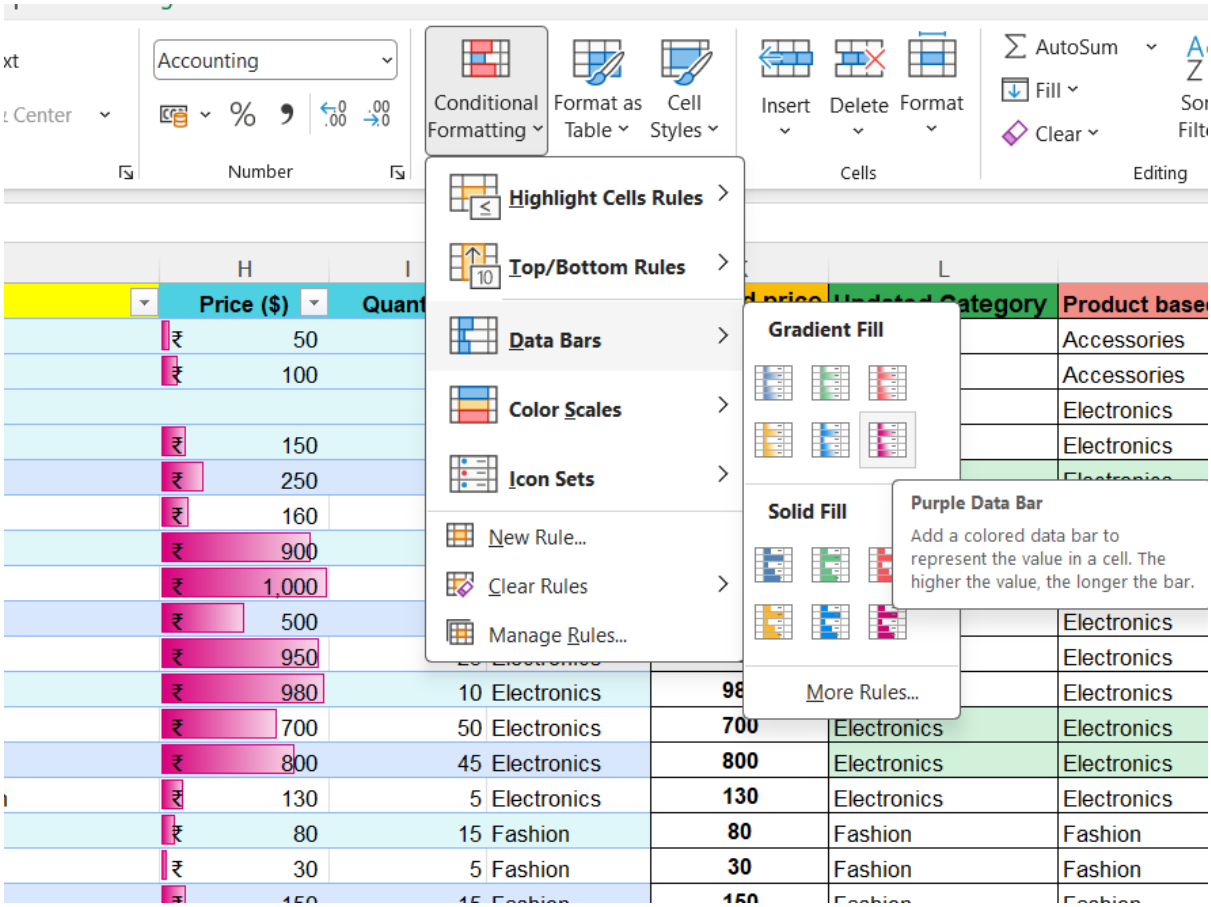



| Product ID | Manufacturing Date | Formated Manufacturing Date |
|---|---|---|
| 21-AUG-CA | 21-AUG | 21-08-2026 |
| 09-JUL-FR | 09-JUL | 09-07-2026 |
| 07-JUN-UK | 07-JUN | 07-06-2026 |
| 18-FEB-CA | 18-FEB | 18-02-2026 |
| 16-APR-ES | 16-APR | 16-04-2026 |
| 20-AUG-CN | 20-AUG | 20-08-2026 |
| 11-APR-US | 11-APR | 11-04-2026 |
| 28-JAN-US | 28-JAN | 28-01-2026 |
| 05-SEP-UK | 05-SEP | 05-09-2026 |
| 17-JUN-IN | 17-JUN | 17-06-2026 |
| 27-JAN-IT | 27-JAN | 27-01-2026 |
| 14-MAY-RU | 14-MAY | 14-05-2026 |
| 29-SEP-CA | 29-SEP | 29-09-2026 |
| 24-MAY-CA | 24-MAY | 24-05-2026 |
| 15-FEB-US | 15-FEB | 15-02-2026 |
| 19-JUL-UK | 19-JUL | 19-07-2026 |

# Conditional Formatting:

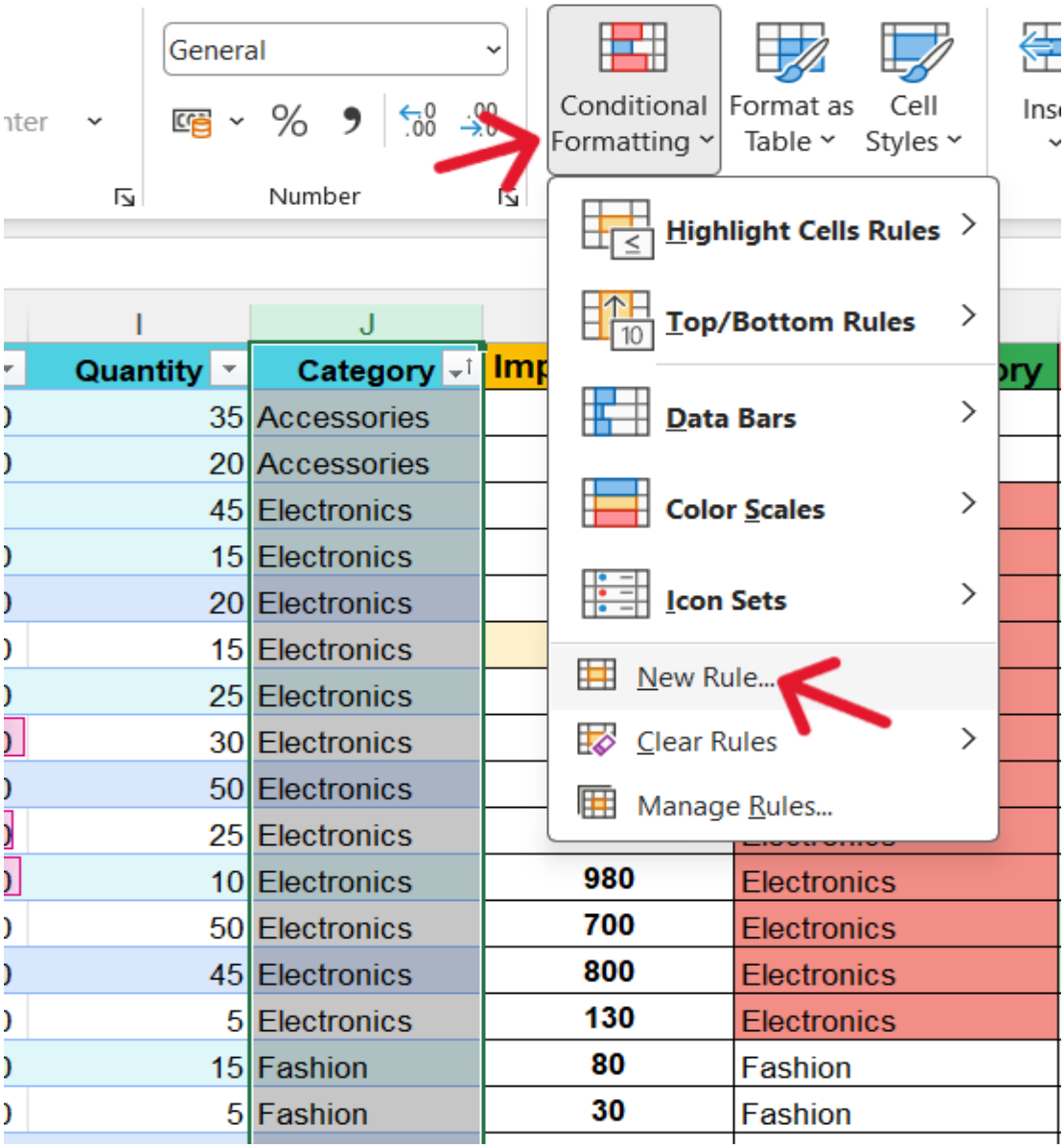
**1. Apply data bar or colour scales conditional formatting in the "Price" column.**

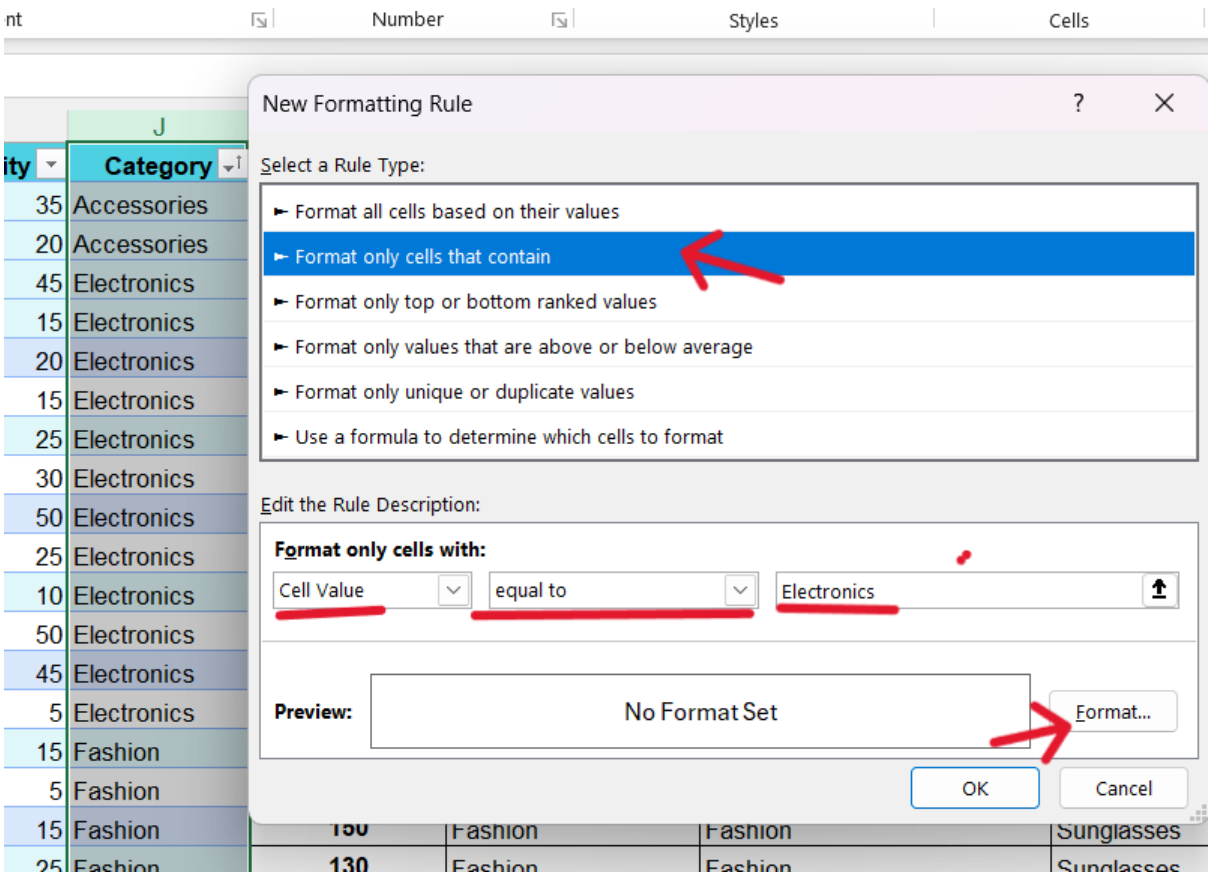
**Applied Data Bars using Conditional Formatting to visually compare values in the Price column**

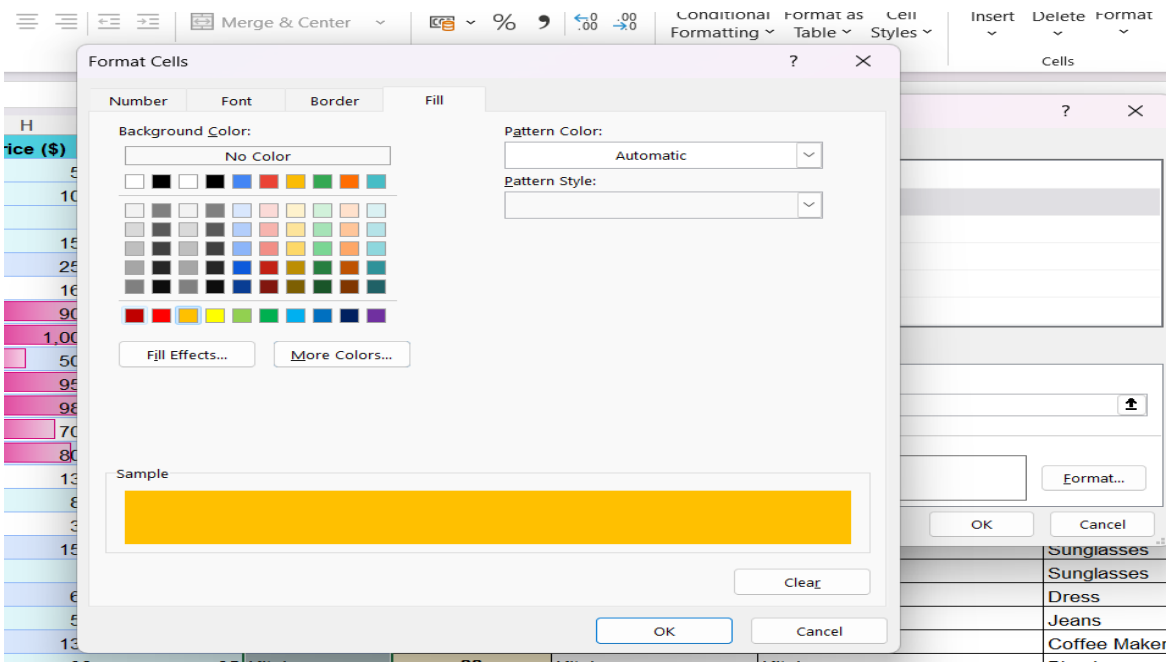


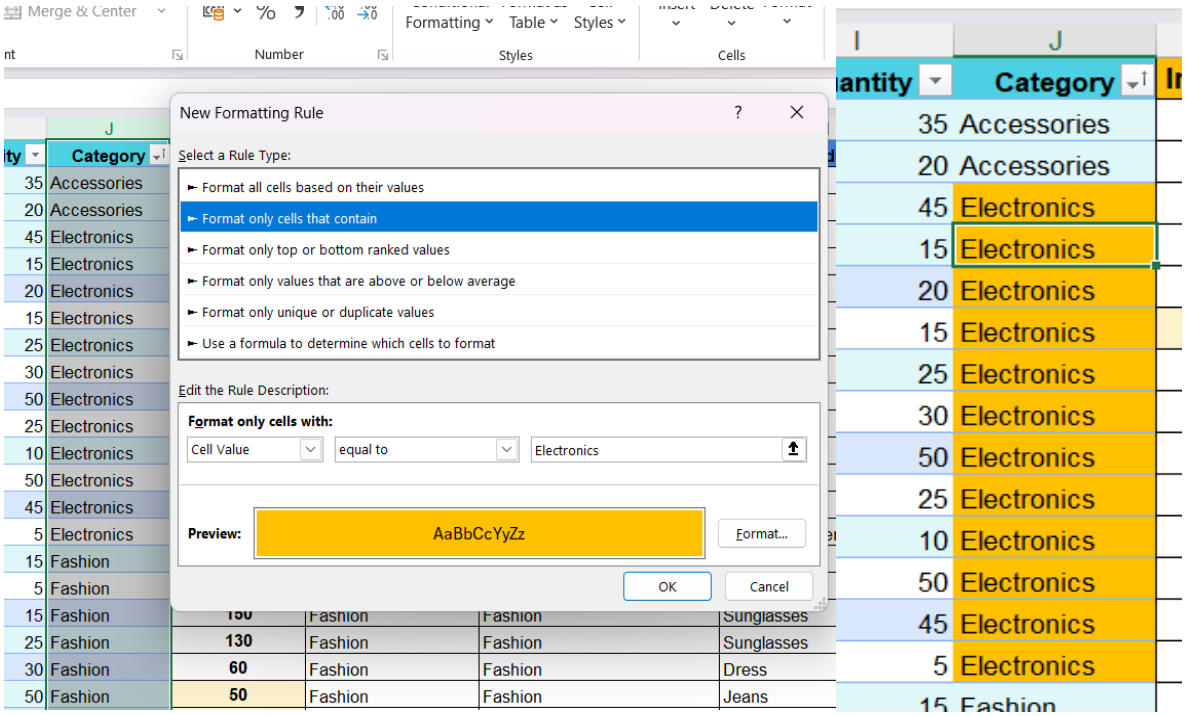
**2. Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."**

**Created a custom Conditional Formatting rule for the Category column using Highlight Cells Rules → Text that Contains, entered 'Electronics', and applied a fill colour to highlight the matching cells.**




Number
Styles
Cells
New Formatting Rule
Select a Rule Type:
Format all cells based on their values
Format only cells that contain
Format only top or bottom ranked values
Format only values that are above or below average
Format only unique or duplicate values
Use a formula to determine which cells to format
Edit the Rule Description:
Format only cells with:
Cell Value
equal to
Electronics
Preview:
No Format Set
Format...
OK
Cancel
Category
Accessories
Electronics
Fashion
Sunglasses


Merge & Center
Conditional Formatting
Format as Table
Cell Styles
Insert
Delete
Format
Cells
Format Cells
Number
Font
Border
Fill
Background Color:
No Color
Pattern Color:
Automatic
Pattern Style:
Fill Effects...
More Colors...
Sample
Clear
OK
Cancel
Format...
OK
Cancel
Sunglasses
Sunglasses
Dress
Jeans
Coffee Maker




## Conclusion:

**Successfully completed data cleaning and formatting using Excel functions and tools. Handled missing values, removed duplicates, corrected spelling mistakes, standardized text, split and merged columns, formatted dates and currency, and applied conditional formatting. The dataset is now clean, organized, and ready for further analysis.**